CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 03



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Recap of Previous Lecture



- Topic: CPU Registers
- Topic: Types of Architecture
- Topic: Program Counter
- Topic: Instruction Register
- Topic: Stack Pointer

Topics to be Covered



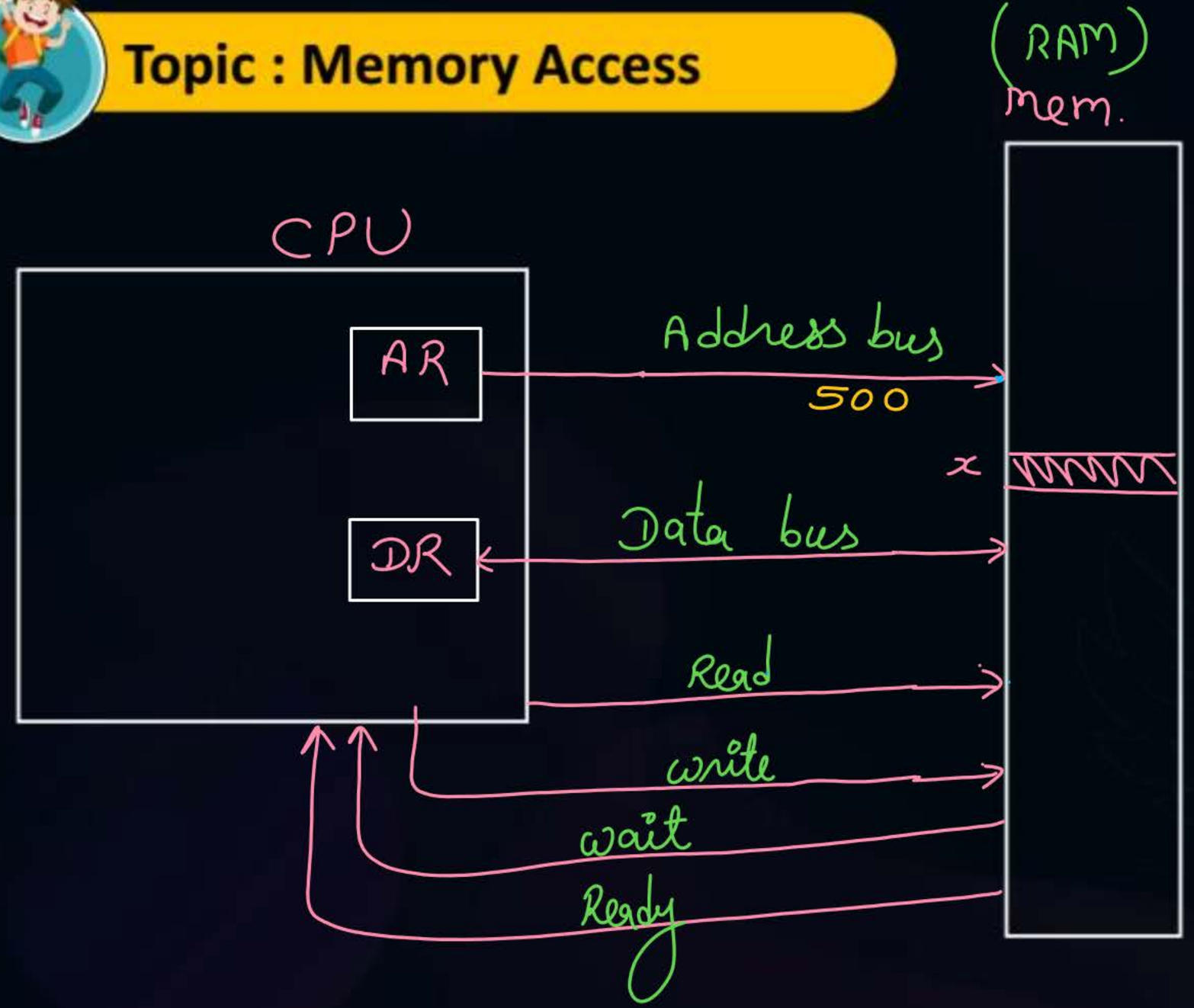
bits ⇒ b
byte ⇒ B

- Topic: Memory Addressing
- Topic: Memory Access
- Topic: Architecture Type (Based on Size of Input)

$\frac{300}{8}$ mbps B



Topic : Memory Access



Mem. operations
→ Read
→ write

Read :-

1. CPU sends address to mem. through address bus.
2. CPU sends enabled memory read control signal
3. memory reads content on given address and sends it to CPU through data bus.

write :-

1. CPU sends address to mem. through address bus
2. CPU sends data to mem. —11— data bus
3. CPU sends enabled mem. write control signal.
4. memory performs write of given content on given address.

when mem is free ⇒ wait ⇒ 0
Ready ⇒ 1

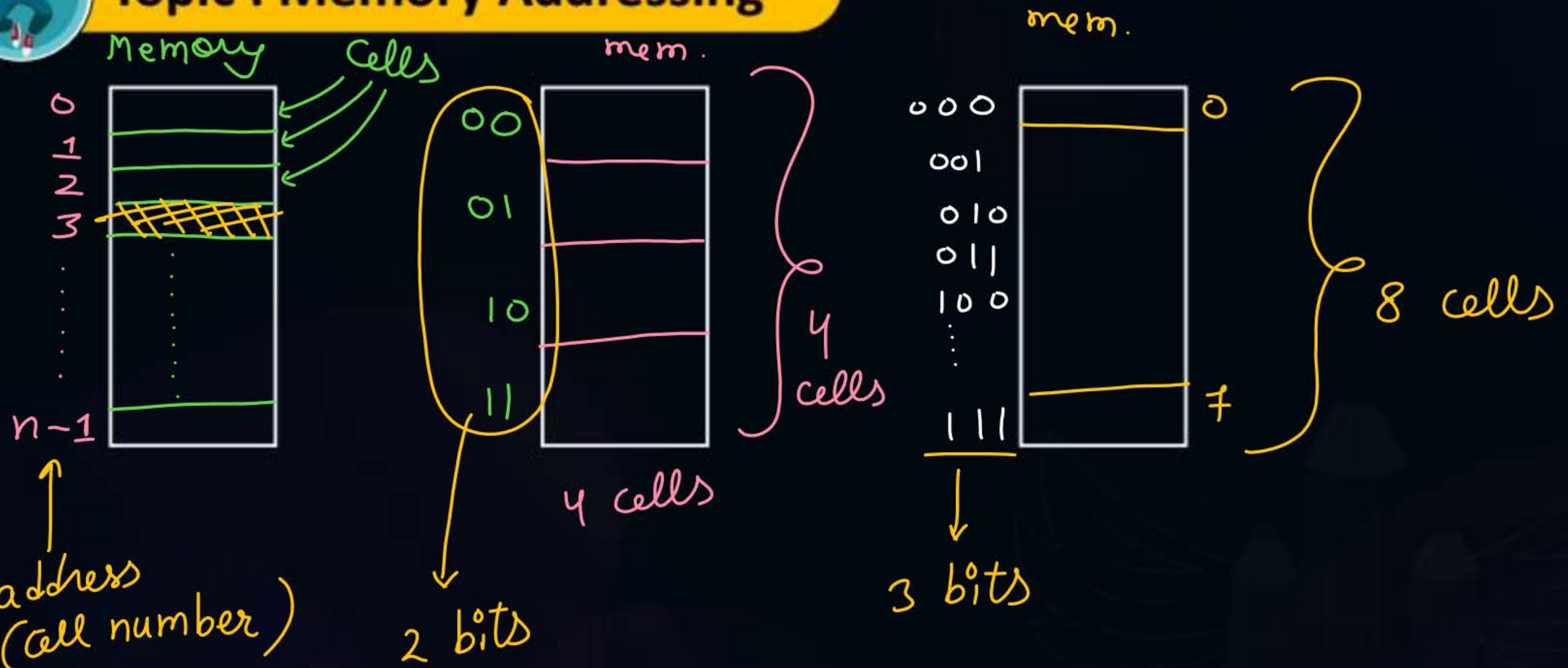
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when mem is performing read or write ⇒ wait ⇒ 1
Ready ⇒ 0

Until mem. gets ready again, CPU can not initiate any operation for it.



Topic : Memory Addressing





Topic : Memory Addressing

no. of cells = 5

add. = $\log_2 5 = 2.\ldots$

no. of cells	$4 = 2^2$	$8 = 2^3$	$16 = 2^4$	2^n	x
address length	2 bits	3 bits	4 bits	n bits	$\lceil \log_2 x \rceil$ bits

floor $\lfloor \quad \rfloor$ $\Rightarrow$ $\lfloor 3.4 \rfloor = 3$ $\lfloor 3.9 \rfloor = 3$ $\lfloor 3.01 \rfloor = 3$

ceil $\lceil \quad \rceil$ $\Rightarrow$ $\lceil 3.4 \rceil = 4$ $\lceil 3.9 \rceil = 4$ $\lceil 3.01 \rceil = 4$

[NAT]

#Q. Number of cells in memory = 2^8

Address size for memory = 8 bits?

[NAT]

#Q. Number of cells in memory = 512 $= 2^9$

Address size for memory = 9 bits?

$2^{10} = 1024 = 1k$

$2^{20} = 1M$

$2^{30} = 1G$

$2^0 = 1$
$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16$
$2^5 = 32$
$2^6 = 64$
$2^7 = 128$
$2^8 = 256$
$2^9 = 512$

$2^{10} = 1024$
$2^{11} = 2048$

[NAT]

#Q. Number of cells in memory = 32M $= 32 * 1M = 2^5 * 2^{20} = 2^{25}$

Address size for memory = 25 bits?

[NAT]

#Q. Number of address bits to access memory = 12 bits

Number of cells in memory = 2^{12}?

$2^{12} = 2^2 * 2^{10} = 4K$

2^{12} → 4096

[NAT]

#Q. Number of address bits to access memory = 27 bits

Number of cells in memory = 2^{27} ? = 128 M

$$2^{27} = 2^{20} * 2^{7}$$

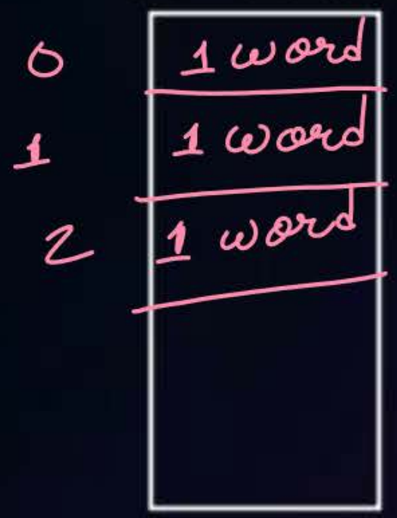
$$= 128\ M$$



Topic : Memory Types

1. Byte Addressable (default) ⇒ each cell of mem. has capacity of 1 byte.
2. Word Addressable ⇒ —11— 11— 1 word.

Word Addressable



1 word ⇓ 1 byte or 2 bytes or 4 bytes or 8 bytes

Byte Addressable



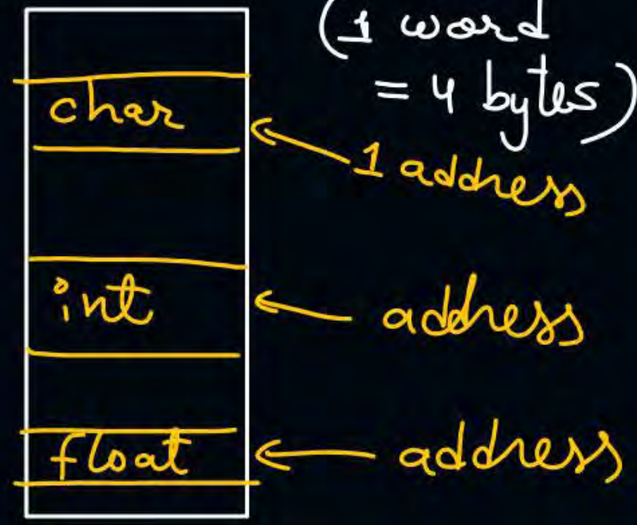
int x;

Assume each int ⇒ 2 bytes

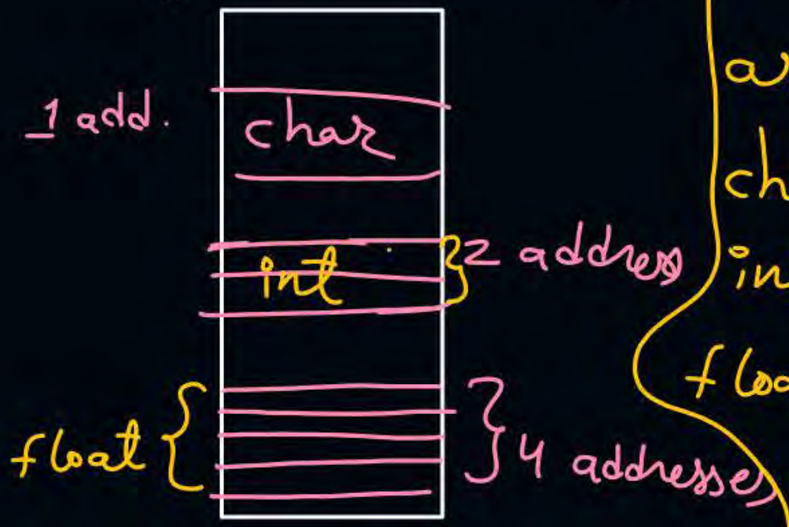
→ to store an 'int' of 2 bytes, 2 addresses of mem. needed

word addressable

(1 word = 4 bytes)



byte addressable



assume,

char ⇒ 1 byte

int ⇒ 2 bytes

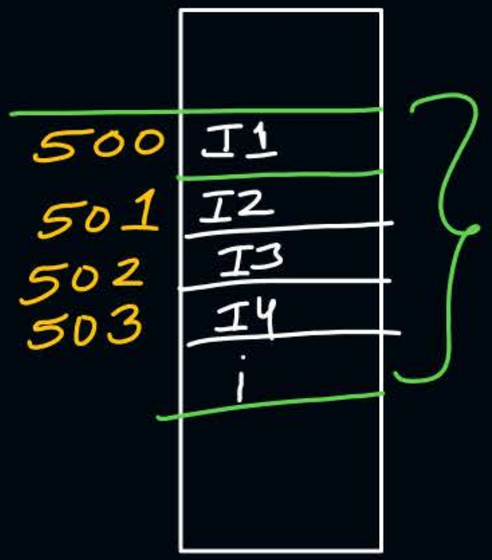
float ⇒ 4 bytes

storing instructions in memory :-

ex ①

Assume each inst^{n} size = 1 byte

byte addressable memory



Program is stored in memory starting from address = 500

what should be add. of instruction I120 ?

$= 500 + (120 - 1)$

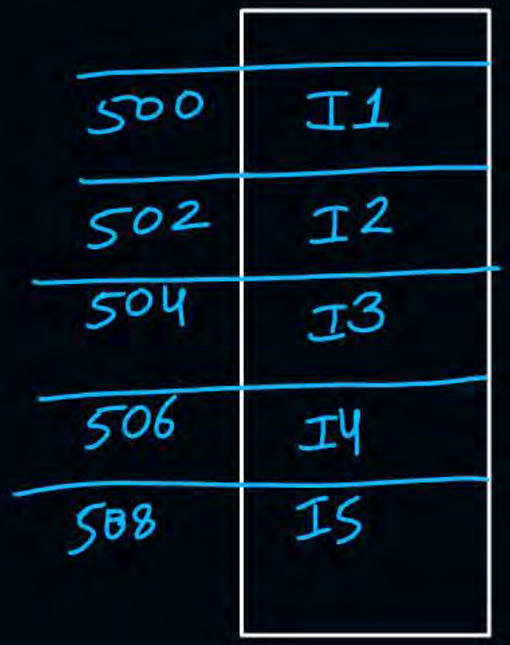
$= 619$

ex :- ②

inst^n size = 2 bytes

mem. byte addressable

Prog. starts in mem. from address = 500



add. of inst^n I120

$= \text{base} + \text{size} * (n-1)$

$= 500 + 2 * (120-1)$

$= 738$

[NAT]

#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 200 (in decimal). Find the address of following instructions:

1. I_1 = 200
2. I_5 = 200 + 4 * (5-1) = 216
3. I_{120} = 200 + 4 * (120-1) = 676

If CPU will have to execute this program then initially PC value will be 200.



PC = ~~200~~ ~~204~~ 208

201, 202, 203 will not be valid PC values

[NAT]

#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 500 (in decimal). What should be the PC value when instruction I_6 will be executing in CPU?

PC = add. of I_7 = 500 + 4 * (7 − 1)

= 524 Ans.

[NAT]

#Q. A CPU has 4 bytes instructions. A program (Instructions I_1 to I_{200}) starts at address 500 (in decimal). What should be the PC value when instruction i will be executing in CPU?

$$PC = \text{add. of } (i+1) = 500 + 4 * (i + 1 - 1)$$

$$= 500 + 4i$$



2 mins Summary

- **Topic** Memory Addressing
- **Topic** Memory Access
- **Topic** Architecture Type (Based on Size of Input)



Happy Learning

THANK - YOU